

Roshjens Toolkit

Data Analysis Toolkit — Project Documentation

https://roshjenskunwar7.github.io/DA_Toolkit/

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7 Modules

15 Statistical
Tests

Works Offline

Any CSV File

No Coding

1. Project Overview

Roshjens Toolkit is a complete, browser-based data analysis platform built as a single HTML file. It requires no installation, no server, no coding, and no internet connection after the initial page load. Any user can upload a CSV file and immediately receive a full statistical analysis with plain-English explanations.

1.1 Live URL

https://roshjenskunwar7.github.io/DA_Toolkit/

Open in any modern browser — Chrome, Firefox, Edge, or Safari. The toolkit loads the built-in IBM HR Employee dataset automatically so it is ready to explore immediately.

1.2 Module Summary

#	Module	What it contains
1	Explorer	Row/column counts, type detection, missing value heatmap, data preview
2	Statistics	Summary table, variable detail + interpretation, frequency tables
3	Charts	Histogram + KDE, box plot, scatter plot, bar chart, correlation matrix
4	Distributions	Interactive PDF/CDF explorer with sliders, Q-Q plot, goodness-of-fit
5	Inference	Full hypothesis test suite — 15 tests with effect sizes
6	Regression	OLS linear regression, R ² , RMSE, residual plot, prediction tool
7	Probability Calc	$P(X \leq x)$, $P(X > x)$, $P(a \leq X \leq b)$, inverse CDF for 5 distributions

2. Quick Start Guide

Getting started takes under one minute. No account, no download, no setup required.

- 1 Open the toolkit**
Visit https://roshjenskunwar7.github.io/DA_Toolkit/ in your browser. The app loads with the built-in employee dataset already active.
- 2 Explore immediately**
Click any tab in the top navigation bar to switch between modules. All charts and tests run on the employee dataset automatically.
- 3 Upload your own data**
Click the Upload CSV button in the top-right corner. A file picker appears. Select any .csv file from your computer. All 7 modules update instantly.
- 4 Read the results**
Every result includes a plain-English explanation. No statistics background is needed to interpret the output.
- 5 Switch datasets**
Click Upload CSV again at any time to load a different file. The entire toolkit resets and re-analyses the new data.

2.1 CSV File Requirements

Requirement	Detail
File format	Must be .csv (comma-separated values)
First row	Must contain column headers (variable names)
Numbers	Do not include currency symbols or commas — use 5000 not \$5,000
Missing values	Handled automatically — filled with column median for numeric columns
File size	Up to ~50,000 rows work well; very large files may be slower
Encoding	UTF-8 recommended

2.2 Tips for Best Results

- Include at least one numeric column and one categorical column to unlock all tests

- A binary column (exactly 2 categories: Yes/No, Male/Female) enables t-tests, proportion tests, and Fisher's exact test
- A multi-category column (3-15 categories: Department, City, Job Role) enables ANOVA and chi-squared tests
- The toolkit automatically chooses parametric or non-parametric tests based on normality

3. Module Reference Guide

Module 1 — Data Explorer

The Explorer loads automatically when you open the toolkit or upload a new file. It gives you an instant overview of your dataset before any analysis begins.

Panel	What it shows
KPI tiles	Total rows, columns, missing cells, number of binary columns
Column Overview	Every column's type (numeric/categorical), non-null count, missing count, unique values
Missing Value Analysis	Colour-coded horizontal bar chart — green = 0% missing, amber = some, red = >10%
Data Preview	First 12 rows of your dataset in a scrollable table

Module 2 — Descriptive Statistics

Computes a complete numerical summary for every variable and generates plain-English interpretations automatically.

Panel	What it shows
Summary Table	Mean, median, std dev, min, Q1, Q3, max, IQR, skewness, kurtosis, CV%, normality flag
Variable Detail	Full breakdown for one selected variable plus auto-generated plain-English description
Frequency Table	Count, percentage, and cumulative % for any column (numeric or categorical)

Term	What it means
Mean	Arithmetic average — affected by outliers
Median	Middle value — robust to outliers; preferred when data is skewed
Std Dev	How spread out values are from the mean
IQR	Interquartile range = Q3 minus Q1 — the spread of the middle 50% of values
Skewness	Positive = long right tail (most values low); Negative = long left tail
Kurtosis	Positive = heavy tails / more outliers; Negative = lighter tails than Normal
CV%	Coefficient of variation = std dev as % of mean — compares spread across variables

Module 3 — Visualisations

Generates the most appropriate charts for your data. Every chart has dropdown menus to select which columns to plot.

Chart	Best used for	Controls
Histogram + KDE	Shape of a numeric variable's distribution	Variable, bins (15/25/40)
Box Plot by Group	Comparing spread across categories	Numeric column, grouping column
Scatter Plot	Relationship between two numeric variables	X-axis, Y axis, optional colour-group
Bar Chart	Frequency of categorical values	Categorical column
Correlation Matrix	Pairwise relationships between all numeric columns	Auto-built (up to 10 cols)

Reading the correlation matrix: Blue cells = positive correlation. Red cells = negative correlation. Darker colour = stronger relationship. Hover over any cell to see the exact r value.

Module 4 — Probability Distributions

Two sub-panels: an interactive distribution explorer with parameter sliders, and a goodness-of-fit section for your actual data.

Distribution Explorer

- Choose a distribution: Normal, Student t, Uniform, Exponential, Beta
- Adjust the two parameter sliders — the PDF and CDF curves update in real time
- PDF (left panel): probability density function — the shape of the distribution
- CDF (right panel): cumulative probability — $P(X \leq x)$ at every point

Goodness-of-Fit Tests

- Select any numeric column from your data
- The Q-Q plot shows your data's quantiles vs the theoretical Normal distribution
- Points close to the red diagonal line = data is approximately Normal
- Three formal tests run automatically: Shapiro-Wilk, Jarque-Bera, KS statistic
- Verdict: 2 out of 3 tests passing = approximately Normal

Module 5 — Statistical Inference

The most comprehensive module. Every test includes the exact H_0 and H_1 for your data, the test statistic and p-value, a plain-English verdict, and an effect size. The Auto Test Engine runs up to 15 tests automatically, selecting parametric or non-parametric methods based on normality.

Auto Hypothesis Test Engine

Reads your columns, generates all sensible test pairs, runs them, and ranks results by significance. Automatically selects:

When data IS Normal	When data is NOT Normal
Welch's t-test	Mann-Whitney U
One-way ANOVA	Kruskal-Wallis
Pearson r	Spearman rho

Full List of Available Tests

Test	H_0 (null hypothesis)	Type
Confidence Interval	Provides range for true population mean	Estimation
One-sample t-test	Population mean equals a specified value	Parametric
Welch two-sample t-test	Two group means are equal	Parametric
Mann-Whitney U	Two group distributions are equal	Non-parametric

Test	H0 (null hypothesis)	Type
One-way ANOVA	All group means are equal (3+ groups)	Parametric
Kruskal-Wallis	All group distributions are equal (3+ groups)	Non-parametric
Paired t-test	Mean of paired differences equals zero	Parametric
Levene's test	All group variances are equal	Variance
Bartlett's test	All group variances are equal (assumes normality)	Variance
Chi-squared independence	Two categorical variables are independent	Categorical
Fisher's exact test	Two binary variables are independent (2x2)	Categorical
One-proportion Z-test	Proportion equals a specified target value	Proportion
Pearson r	No linear correlation between two variables	Correlation
Spearman rho	No rank correlation between two variables	Correlation

Module 6 — Linear Regression

Fits an ordinary least squares (OLS) regression model to predict one numeric variable from another.

Panel	What it shows
Model Setup	Intercept (a), slope (b), R ² , RMSE, slope t-statistic and p-value
Scatter + Regression Line	Data points with the fitted line overlaid in green
Residuals vs Fitted	Diagnostic plot — checks whether model assumptions are met
Predict	Enter any x value and instantly see the predicted y

Reading the residual plot: Random scatter around zero = good fit. Funnel shape = heteroscedasticity. Curved pattern = non-linear relationship.

Module 7 — Probability Calculator

Computes exact probabilities for any of five distributions: Normal, Student t, Uniform, Exponential, and Beta.

Query type	What it computes
$P(X \leq x)$	Probability of a value being at most x (left-tail area)
$P(X \geq x)$	Probability of a value being at least x (right-tail area)
$P(a \leq X \leq b)$	Probability of a value falling between a and b
Inverse CDF	Find x such that $P(X \leq x) = p$ (critical values, percentiles)

Common use cases:

- What is the probability of a standard normal value above 1.96? -> Normal, $\mu=0$, $\sigma=1$, $P(X \geq x)$, $x=1.96$
- What t-value cuts off the top 5% with 29 df? -> t distribution, $df=29$, Inverse CDF, $p=0.95$
- What proportion falls between 50 and 70 in $N(60, 10)$? -> Normal, $\mu=60$, $\sigma=10$, $P(a \leq X \leq b)$, $a=50$, $b=70$

4. Interpreting Results

4.1 Understanding p-values

The p-value is used in every hypothesis test. It is the probability of seeing results as extreme as yours if the null hypothesis were actually true. A small p-value means the result is unlikely to be due to chance alone.

p-value	Evidence	Decision
$p < 0.001$	Very strong against H0	Reject H0
$0.001 \leq p < 0.01$	Strong against H0	Reject H0
$0.01 \leq p < 0.05$	Moderate — statistically significant	Reject H0
$0.05 \leq p < 0.10$	Weak — marginally significant	Do not reject
$p \geq 0.10$	Insufficient evidence	Do not reject H0

Important: A significant p-value means the result is unlikely due to chance. It does NOT mean the effect is large or practically important. Always check the effect size alongside the p-value.

4.2 Effect Size Reference

Effect sizes tell you how big a difference is in practical terms, not just whether it is statistically significant.

Measure	Used in	Negligible	Small	Medium	Large
Cohen's d	t-tests	< 0.2	0.2	0.5	> 0.8
Eta-sq (η^2)	ANOVA	< 0.01	0.01	0.06	> 0.14
Cramer's V	Chi-sq, Fisher	< 0.1	0.1	0.3	> 0.5
r / rho	Correlation	< 0.1	0.1	0.3	> 0.5

4.3 Normality and Test Selection

Many statistical tests assume the data follows a Normal distribution. The toolkit checks this automatically using Shapiro-Wilk and Jarque-Bera tests and switches to the appropriate test.

Situation	Parametric test	Non-parametric alternative
Comparing 2 group means	Welch t-test	Mann-Whitney U
Comparing 3+ group means	One-way ANOVA	Kruskal-Wallis

Situation	Parametric test	Non-parametric alternative
Measuring linear relationship	Pearson r	Spearman rho

5. Troubleshooting

Problem	Solution
Charts not showing / page blank	Make sure JavaScript is enabled in your browser. Try Chrome or Firefox.
Upload button does nothing	Try a different browser. Safari on older macOS may restrict local file access.
CSV not loading correctly	Check first row has column headers and values are comma-separated (not semicolons).
Numbers showing as categories	Remove currency symbols (\$), commas in numbers, or percent signs (%) from the CSV.
Tests showing 'Need 3+ groups'	Your grouping column has fewer than 3 categories with enough observations.
All variables show Non-Normal	Common for real-world data. The toolkit automatically uses non-parametric tests.
Correlation matrix is blank	Need at least 2 numeric columns in your dataset.
Fisher's exact shows no results	Requires two columns each with exactly 2 unique values (binary columns).
Regression shows no line	Select two different numeric columns — predictor and outcome cannot be the same.
Page loads but nothing works	Hard refresh with Ctrl+Shift+R (Windows) or Cmd+Shift+R (Mac).

6. Project Submission Details

Item	Detail
Live URL	https://roshjenskunwar7.github.io/DA_Toolkit/
Platform	GitHub Pages (static hosting)
File	index.html — single self-contained file
Dataset	IBM HR Employee dataset — 1,470 rows x 31 columns
Tests	15 hypothesis tests with effect sizes
Modules	7 (Explorer, Statistics, Charts, Distributions, Inference, Regression, Probability)
Coding required	None — fully no-code interface

